SAG - P01558 - DALI lot 2 nPFS - offers v2 [IN PROGRESS]

s dali

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Introduction

Document objectives

This document aims to describe the global architecture planned for the introduction of a mediation layer in the context of nPFS project. It will especially define the interactions of the new mediation layer with the different technical components involved in the suggested solution.

Related documents

- DALI swagger: DALI offer management v2 swagger (TB3)
- SAG nPFS Souscription offres (before DALI): [SAG][DROP2&3] Souscription offres (OLD - Sans DALI)
- SI swagger: https://gitlab.int.nbyt.fr/CatalogueInterfaces/referentiel/-/tree/NPFS_Lot2/apis

Context and scope

nPFS project context

nPFS project consists mainly of replacing the current TV platforms (SmartVision, Commscope CMS) with a new one prvided by Kaltura. This therefore implied a new integration for the different device applications that must communicate with the new platform.

Offer management v2 via DALI objectives

This new version of offer API delivered by DALI aims to improve the first version with:

- Support of unmanaged applications (e.g. Canal+) which are not known by DSI

- Add commercial data to offer types: APPLICATION, MULTIOFFER, SERVICE
- Support offers with "API partenaires" (e.g Dorcel and Disney+)

Functional requirements

- **REQ-FUN-01**
  The device shall be able to retrievel all entitled subscriptions with all related metadata (description, images, ..) and commercial details.
  This data is intended to be stored in local device data base.

- **REQ-FUN-02**
  The device shall be able to retrieve all subscribable subscriptions with all related metadata (description, images, ..) and commercial details.
  This data is intended to be stored in local device data base.

- **REQ-FUN-03**
  For each subscription, the device shall be able to display the list of linear channels, Replay and related sVoD catalogs included.
  EPG to be added

- **REQ-FUN-04**
  The end-user shall be able to subscribe to a standard offer through his device UI.
  a standard offer is an offer for which subscription process is managed only by Bouygues SI.

- **REQ-FUN-05**
  The end-user shall be able to subscribe to an offer linked to a partner application with inbox account creation from his device (e.g. Dorcel).
  The end-user shall be able to activate his partner account using the partner application.
  This type of offer needs the account creation on partner back-end before final activation in Bouygues SI.

- **REQ-FUN-06**
  The end-user shall be able to subscribe to an offer linked to a partner application with account creation in partner website through Bouygues customer portal.(e.g. Disney+)
  The end-user shall be able to activate his partner account using the relevent QR code displayed in the UI.
  For this type of offer, subscription is validated by Bouygues SI as soon as the subscribtion is requested. But the access to the service requires final activation after account creation on partner side.

- **REQ-FUN-07**
  The end-user shall be able to unsubscribe any type of offer through his device UI.

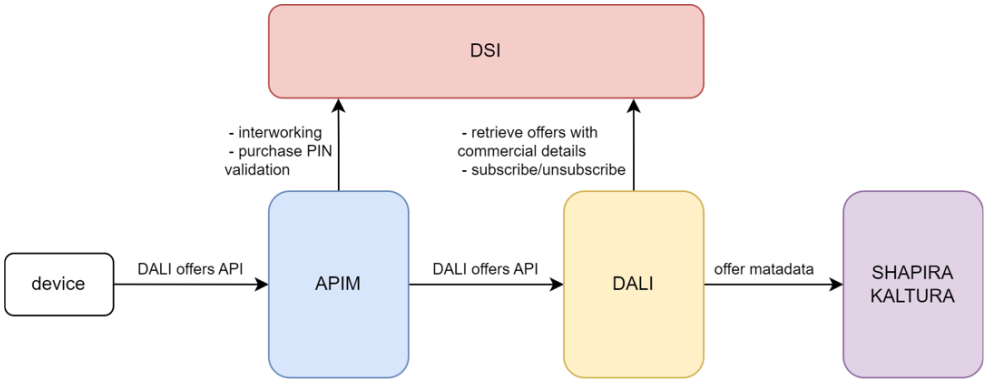
- **REQ-FUN-08**
  The end-user shall be able to have access to the details of a specific offer.
  This specific offer is selected by the user from an editorial row in the UI.
- **REQ-FUN-09**
  The subscription and unsubscription processes shall be protected by the purchase PIN validation.
- **REQ-FUN-10**
  The end-user shall be able to have access to the details of a specific offer.
  This specific offer is selected by the user from an editorial row in the UI.

Subscription retrieval triggers

Every STB will trigger subscriptions retrieval process (both subscribable and subscribed) **every 6 hours**. In order to avoid an overload, the offer queries will be randomely spread by STB application over a 6 hours window.

High level design

Global overview



Technical solution

Offer data model and mapping

To be updated with the final offer modelization.

ℹ DALI swagger is up to date regarding the modelization and shall be used as reference.

Get offers

This service allows the user to retrieve all available offers: either already subscribed, subscribable or both of them.

API

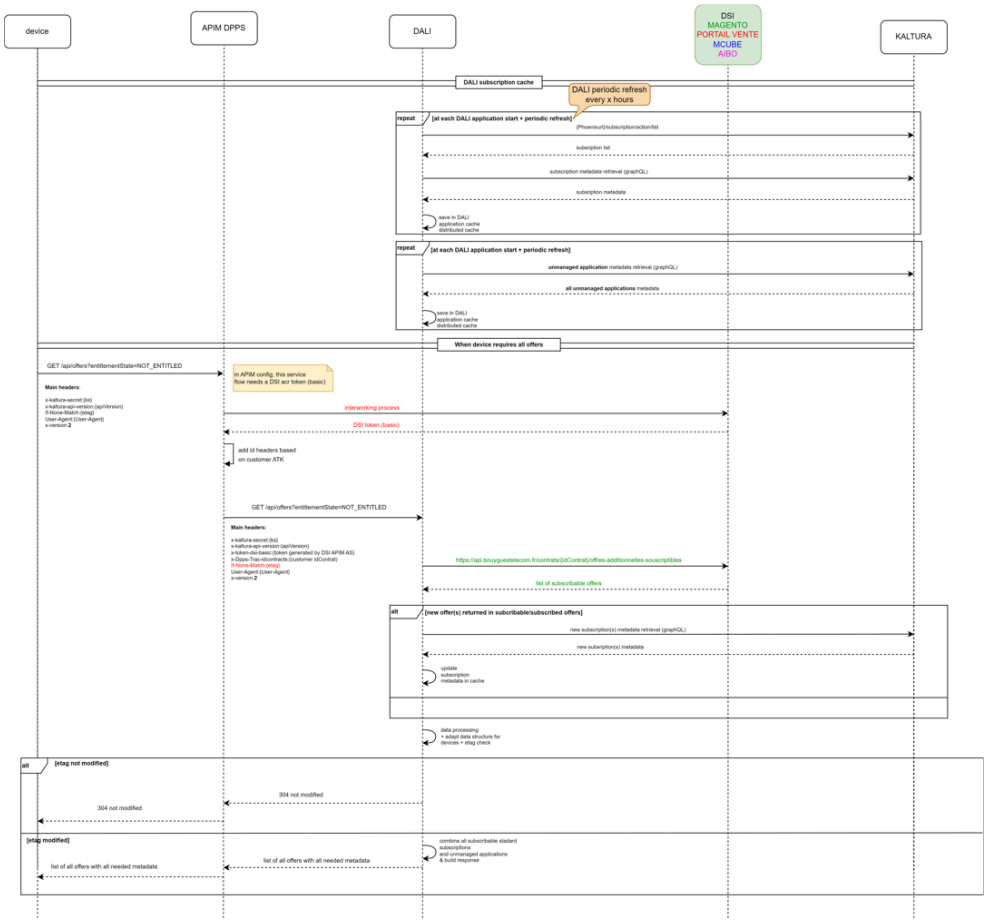
ℹ DALI swagger remains the reference for API details (errors included). This table aims only to explain the use of some parameters.

Input	Description / comments	Type
x-kaltura-secret *required* *(header)*	Kaltura secret retrieved during authent process. This is required for the identification of the customer/device on Kaltura platform	string
x-kaltura-api-version *required* *(header)*	Katura API version.	string
entitlementState *optional* *(query param)*	possible values : SUBSCRIBABLE, SUBSCRIBED, ALL. If not filled → ALL Only offers with the indicated status will be returned.	string
categories *optional* *(query param)*	Offer item categories to retrieve. Possible values: "BOUQUET", "APPLICATION", "SERVICE", "MULTIOFFER", "UNMANAGED_APPLICATION" If not filled → ALL	string
fields *optional* *(query param)*	list of attributes to be returned in the offer object.	Array of strings
x-version *optional* *(header)*	DALI offer API version	string

Callflows

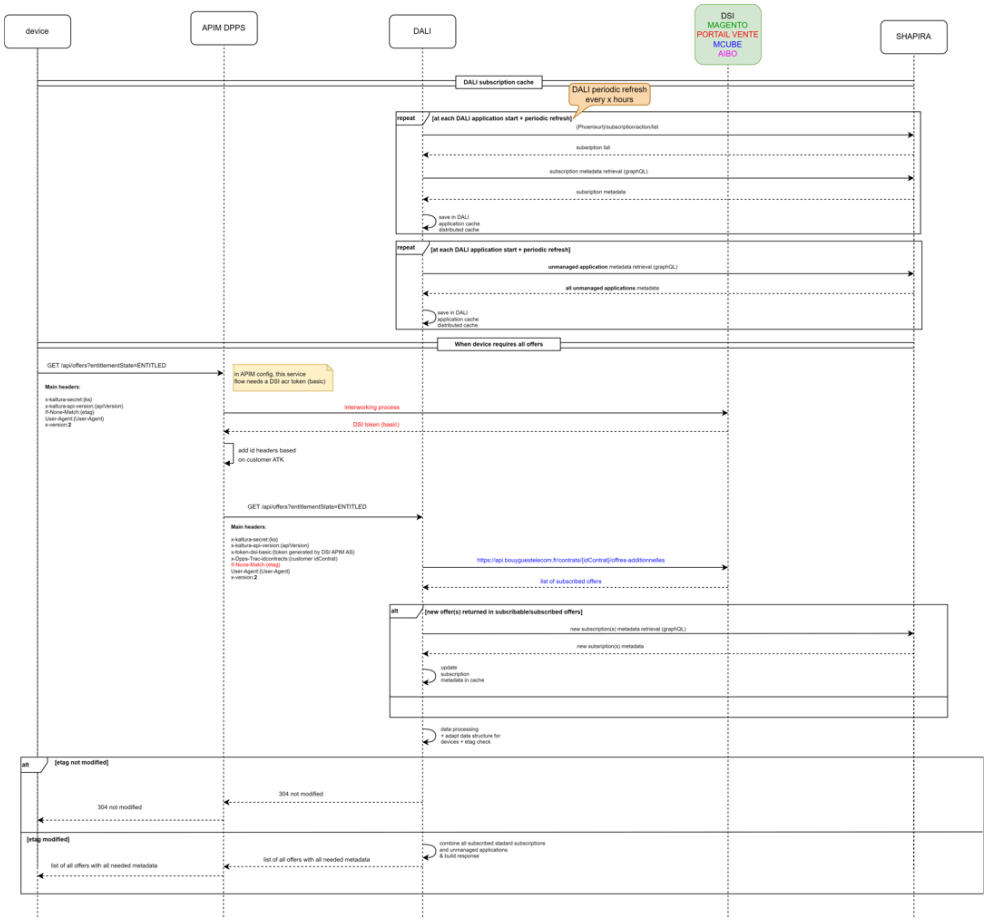
ℹ Intercation with DSI platform will be detailed later if needed.

Subscribable offers

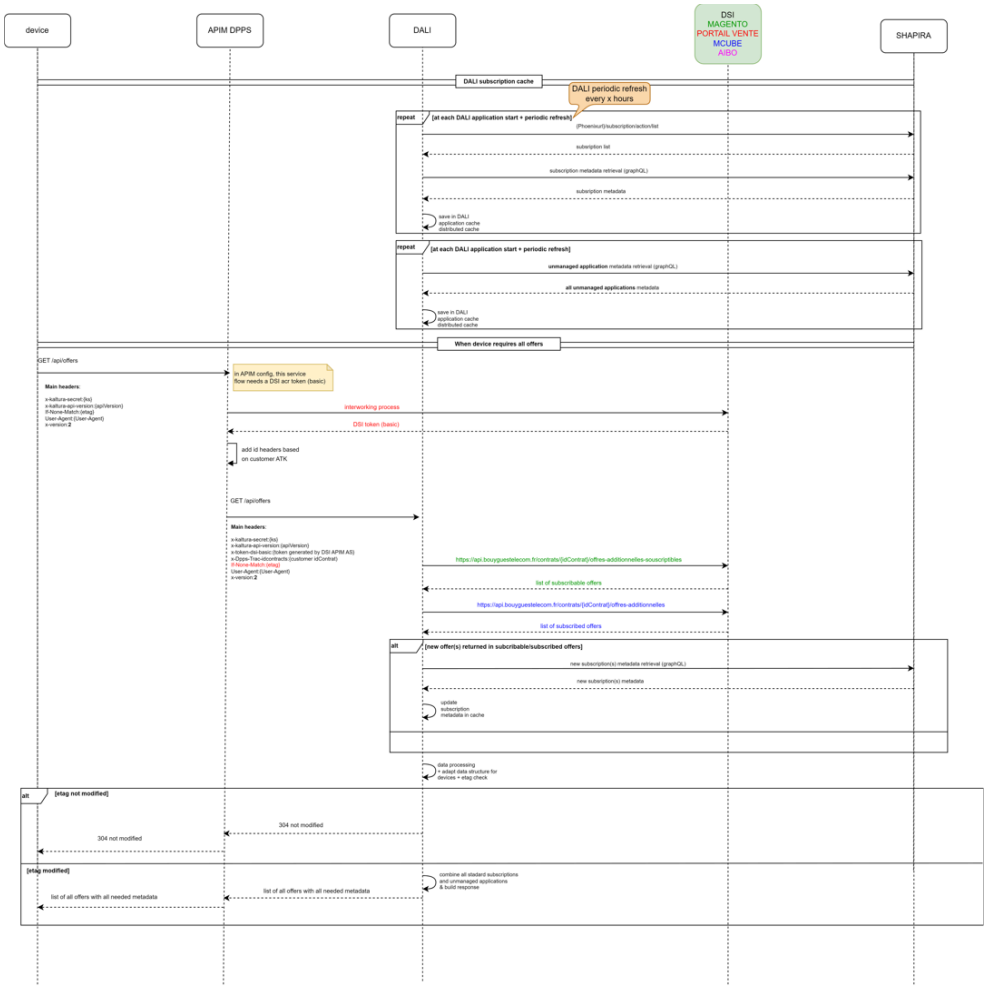


Subscribed offers

After alignement with DSI teams, it has been decided to always retrieve subscribed offers from DSI. Dimensionning questions have been taken into account by DSI teams.



All offers (subscribed and unsubscribed)



Errors

HTTP Code	Description	Example

400	**Bad Request (with error** `=` Bad Request**)**	{
		• "error": "Bad Request",
		• "errorDescription": "Invalid format",
		• "errorSource": "string"
		}

401	**INVALID_TOKEN :** Invalid Token	{
		• "error": "Invalid Token",
		• "errorDescription": "Token is invalid",
		• "errorSource": "string"
		}

403	**Forbidden** **Possible** `resultMessage` **values :** **- INVALID_TOKEN : Token Expired**	

404	**Not Found**
	Possible `resultMessage` **values:**
	- `CUSTOMER_NOT_FOUND :` Customer Not Found
		{
		• "error": "Customer Not Found",
		• "errorDescription": "Customer wasn't found",
		• "errorSource": "string"
		}

405	**Method Not Allowed** Possible error value : Method Not Allowed	{ • "error": "Method Not Allowed",

		• "errorDescription": "This HTTP Method is not available on this service", • "errorSource": "string" }

500	**Internal error (with error `INTERNAL_SERVER_ERROR` : Internal Server Error)**	{
		• "error": "Internal Server Error",
		• "errorDescription": "Internal Server Error",
		• "errorSource": "string"
		}

502	**Internal error (with error `EXTERNAL_SERVER_ERROR` : Internal Server Error)**	{
		• "error": "External Server Error",
		• "errorDescription": "External Server Error",
		• "errorSource": "string"
		}

Subscribe

This service allows the user to subscribe to a subscribable offer.

API

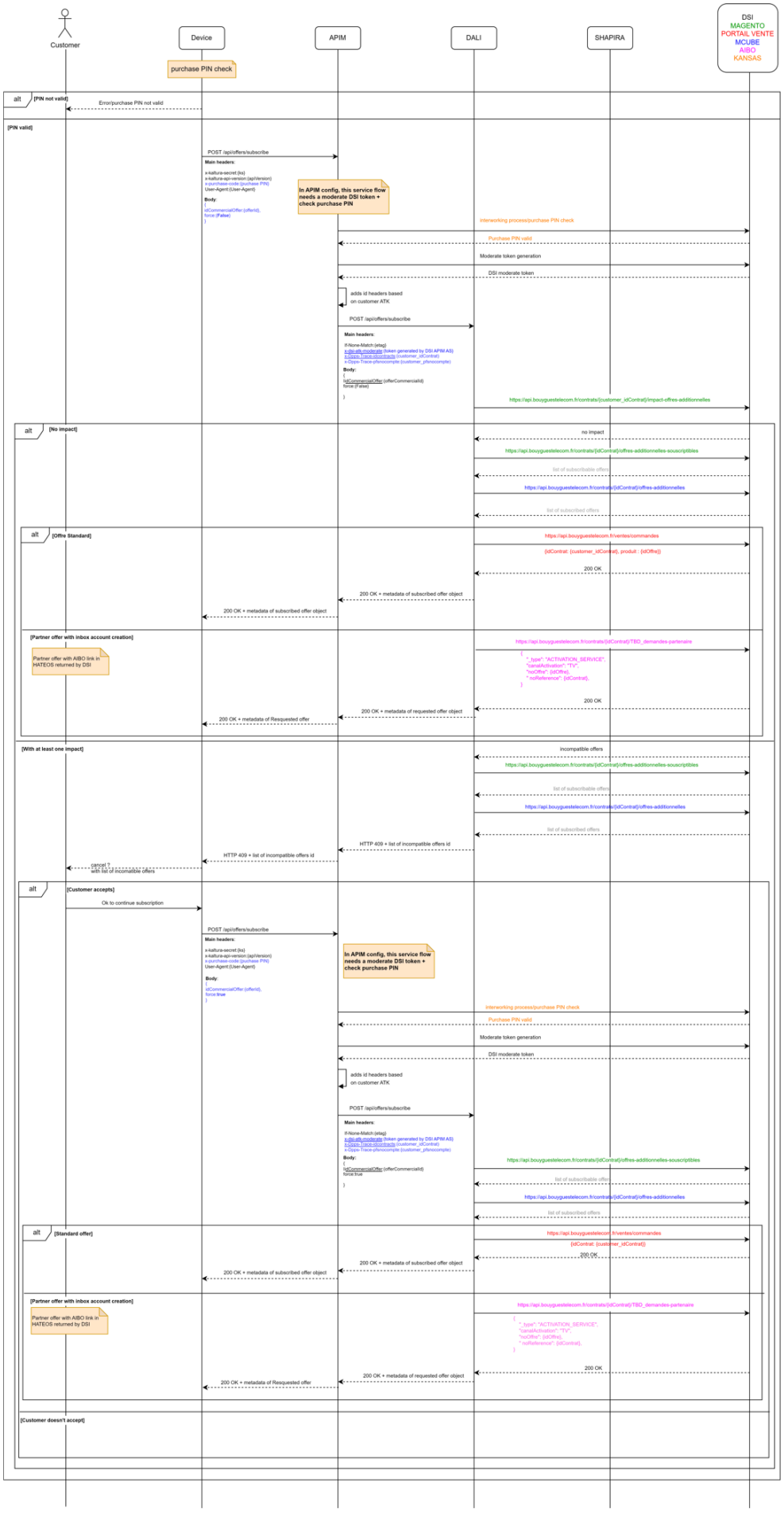
ℹ DALI swagger remains the reference for API details (errors included). This table aims only to explain the use of some parameters.

```
POST /api/offers/subscribe
```

Input	Description / comments	Type
x-kaltura-secret *required* *(header)*	Kaltura secret retrieved during authent process. This is required for the identification of the customer/device on Kaltura platform	string
x-kaltura-api-version *required* *(header)*	Katura API version.	string
x-purchase-code *required* *(header)*	purchase PIN	string
idCommercialOffer *required* *(body)*	commercial id of the offer that will be used to call DSI API.	string
force *optional* *(body)*	Flag used by the device to indicate that the subscription shall be processed even if there is a incompatible offer.	boolean
x-version *optional* *(header)*	DALI offer API version	string

Callflow

ℹ Intercation with DSI platform will be detailed later if needed.


Customer
Device
APIM
DALI
SHAPIRA
DSI
MAGENTO
PORTAIL VENTE
MCUBE
AIBO
KANSAS
purchase PIN check
alt
[PIN not valid]
Error/purchase PIN not valid
[PIN valid]
POST /api/offers/subscribe
Main headers:
x-kaltura-secret: (ks)
x-kaltura-api-version:(apiVersion)
x-purchase-code:(purchase PIN)
User-Agent:(User-Agent)
Body:
{
idCommercialOffer:(offerId),
force:(False)
}
In APIM config, this service flow needs a moderate DSI token + check purchase PIN
interworking process/purchase PIN check
Purchase PIN valid
Moderate token generation
DSI moderate token
adds id headers based on customer ATK
POST /api/offers/subscribe
Main headers:
If-None-Match:(etag)
x-dsi-atk-moderate (token generated by DSI APIM AS)
x-Dsps-Trace-idcontracts (customer_idContrat)
x-Dsps-Trace-pfsnocompte:(customer_pfsnocompte)
Body:
{
idCommercialOffer: (offerCommercialId)
force:(False)
}
https://api.bouyguestelecom.fr/contrats/{customer_idContrat}/impact-offres-additionnelles
[No impact]
no impact
https://api.bouyguestelecom.fr/contrats/{idContrat}/offres-additionnelles-souscriptibles
list of subscribable offers
https://api.bouyguestelecom.fr/contrats/{idContrat}/offres-additionnelles
list of subscribed offers
[Offre Standard]
https://api.bouyguestelecom.fr/ventes/commandes
{idContrat: {customer_idContrat}, produit : {idOffre}}
200 OK
200 OK + metadata of subscribed offer object
200 OK + metadata of subscribed offer object
[Partner offer with inbox account creation]
Partner offer with AIBO link in HATEOS returned by DSI
https://api.bouyguestelecom.fr/contrats/{idContrat}/TBD_demandes-partenaire
{
"_type": "ACTIVATION_SERVICE",
"canalActivation": "TV",
"noOffre": {idOffre},
" noReference": {idContrat},
}
200 OK
200 OK + metadata of requested offer object
200 OK + metadata of Resquested offer
[With at least one impact]
incompatible offers
https://api.bouyguestelecom.fr/contrats/{idContrat}/offres-additionnelles-souscriptibles
list of subscribable offers
https://api.bouyguestelecom.fr/contrats/{idContrat}/offres-additionnelles
list of subscribed offers
HTTP 409 + list of incompatible offers id
HTTP 409 + list of incompatible offers id
cancel ? with list of incompatible offers
[Customer accepts]
Ok to continue subscription
POST /api/offers/subscribe
Main headers:
x-kaltura-secret: (ks)
x-kaltura-api-version:(apiVersion)
x-purchase-code:(purchase PIN)
User-Agent:(User-Agent)
Body:
{
idCommercialOffer:(offerId),
force:true
}
In APIM config, this service flow needs a moderate DSI token + check purchase PIN
interworking process/purchase PIN check
Purchase PIN valid
Moderate token generation
DSI moderate token
adds id headers based on customer ATK
POST /api/offers/subscribe
Main headers:
If-None-Match:(etag)
x-dsi-atk-moderate (token generated by DSI APIM AS)
x-Dsps-Trace-idcontracts (customer_idContrat)
x-Dsps-Trace-pfsnocompte:(customer_pfsnocompte)
Body:
{
idCommercialOffer: (offerCommercialId)
force:true
}
https://api.bouyguestelecom.fr/contrats/{idContrat}/offres-additionnelles-souscriptibles
list of subscribable offers
https://api.bouyguestelecom.fr/contrats/{idContrat}/offres-additionnelles
list of subscribed offers
[Standard offer]
https://api.bouyguestelecom.fr/ventes/commandes
{idContrat: {customer_idContrat}}
200 OK
200 OK + metadata of subscribed offer object
200 OK + metadata of subscribed offer object
[Partner offer with inbox account creation]
Partner offer with AIBO link in HATEOS returned by DSI
https://api.bouyguestelecom.fr/contrats/{idContrat}/TBD_demandes-partenaire
{
"_type": "ACTIVATION_SERVICE",
"canalActivation": "TV",
"noOffre": {idOffre},
" noReference": {idContrat},
}
200 OK
200 OK + metadata of requested offer object
200 OK + metadata of Resquested offer
[Customer doesn't accept]

Errors

HTTP Code	Description	Example

400	**Bad Request (with error `=` Bad Request)**	{
		• "error": "Bad Request",
		• "errorDescription": "Invalid format",
		• "errorSource": "string"
		}

401	**INVALID_TOKEN :** Invalid Token	{
		• "error": "Invalid Token",
		• "errorDescription": "Token is invalid",
		• "errorSource": "string"
		}

403	**Forbidden** **Possible `resultMessage` values :** - Customer Suspended - Unable Purchase Free - Unable Purchase Offer Purchased	
404	**Not Found** **Possible `resultMessage` values:** - Customer Not Found - Purchase Failed - Offer Not Found	

405	**Method Not Allowed**
	Possible error value : Method Not Allowed
		{
		• "error": "Method Not Allowed",
		• "errorDescription": "This HTTP Method is not available on this service",
		• "errorSource": "string"
		}

409	**Conflict** Possible error value : Incompatible offer	

500	**Internal error (with error `INTERNAL_SERVER_ERROR :` Internal Server Error)**	{
		• "error": "Internal Server Error",
		• "errorDescription": "Internal Server Error",
		• "errorSource": "string"
		}

502	**Internal error (with error `EXTERNAL_SERVER_ERROR :` Internal Server Error)**	{
		• "error": "External Server Error",
		• "errorDescription": "External Server Error",
		• "errorSource": "string"
		}

Partner account activation

Depending on the subscription type (metadata retrieved from SHAPIRA platform), the parter account activation will be triggered either using a QR code or using the partner application. The distinction will be managed by the subscription type metadata and the finalActivationLink returned by DALI to the device during subscribe process.

The finalActivationLink will be built by DALI as follows:

- if subscription type is 2 (e.g. Disney+ use case): finalActivationLink = QRCODE_ACTIVATION_LINK+ commercialOfferId
  QRCODE_ACTIVATION_LINK is configured on DALI side (given by DSI).

- if subscription type is 3 (e.g Dorcel use case): finalActivationLink = activationLink+idServicePartenaire
  activationLink: metadata retrieved from SHAPIRA platform
  idServicePartenaire: info retrieved from DSI response (impactOffresAdditionnelles call)

Unsubscribe

This service allows the user to unsubscribe a subscribed offer.

API

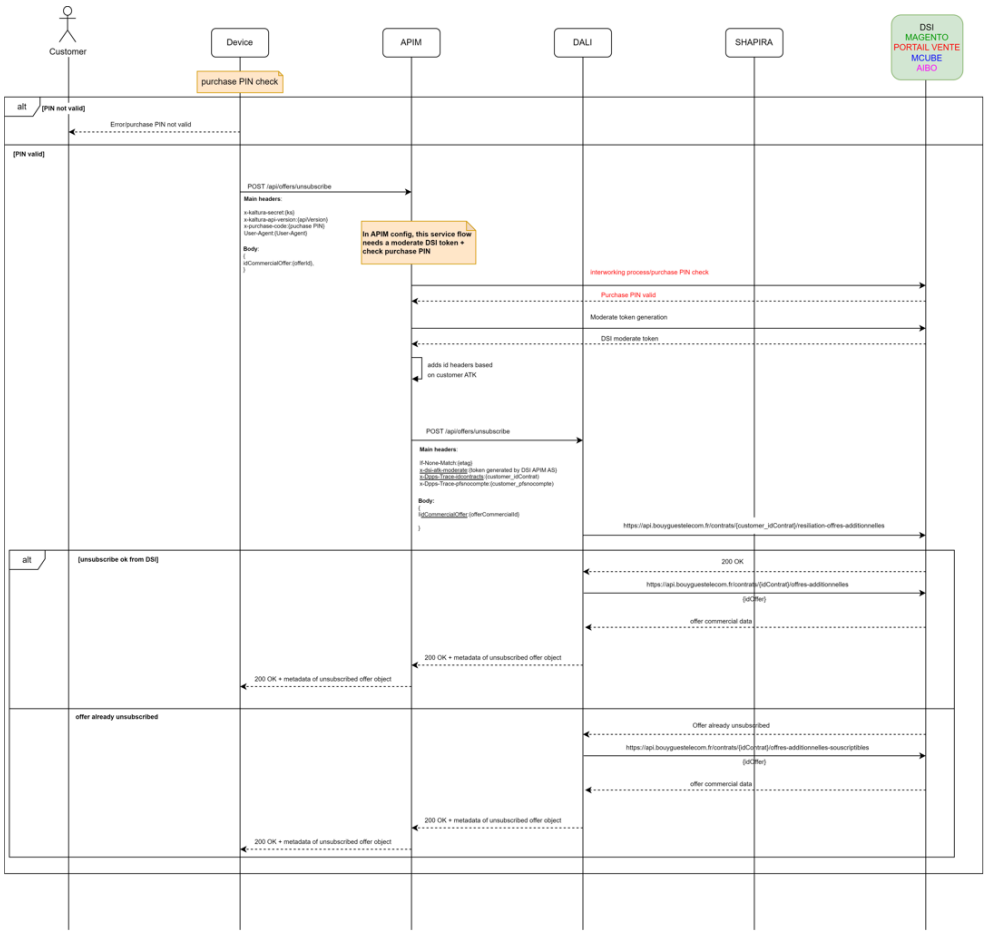
ℹ DALI swagger remains the reference for API details (errors included). This table aims only to explain the use of some parameters.

```
POST /api/offers/unsubscribe
```

Input	Description / comments	Type
x-kaltura-secret *required* *(header)*	Kaltura secret retrieved during authent process. This is required for the identification of the customer/device on Kaltura platform	string
x-kaltura-api-version *required* *(header)*	Katura API version.	string
x-purchase-code *required* *(header)*	purchase PIN	string
idCommercialOffer *required* *(body)*	commercial id of the offer that will be used to call DSI API.	string
x-version *optional* *(header)*	DALI offer API version	string

Callflow

ℹ Intercation with DSI platform will be detailed later if needed.



Errors

HTTP Code	Description	Example

400	**Bad Request (with error `=` Bad Request)**	{
		• "error": "Bad Request",
		• "errorDescription": "Invalid format",
		• "errorSource": "string"
		}

401	**INVALID_TOKEN :** Invalid Token	{
		• "error": "Invalid Token",
		• "errorDescription": "Token is invalid",
		• "errorSource": "string"
		}

403	**Forbidden** **Possible `resultMessage` values :**	
404	**Not Found** **Possible `resultMessage` values:** - Customer Not Found	

500	**Internal error (with error `INTERNAL_SERVER_ERROR` : Internal Server Error)**	{
		• "error": "Internal Server Error",
		• "errorDescription": "Internal Server Error",
		• "errorSource": "string"
		}

502	**Internal error (with error `EXTERNAL_SERVER_ERROR` : Internal Server Error)**	{

		• "error": "External Server Error", • "errorDescription": "External Server Error", • "errorSource": "string" }

Open questions

DSI related questions

- [x] what are the limitations on DSI side to use DSI API for subscribed offers also ?

Here are details agreed with DSI teams (see comments from FG Filipe GOMES)

Ressource/Service	Exposition	ST Source	Temps Interne (hors appel externe)	Temps global	Nombre d'appels/s supplémentaires
grappes-contrats grappes-formules-souscrites grappes-historiques-formules-souscrites	REFCLI	M3 Offre	20ms	30ms	~~100 en nominal / 250 en Pic~~ 400 en nominal / 500 en Pic
rechercherContexteClientIdentite	REFCLI	FLORIDA AS DSI	20ms	50ms	~~200 en nominal / 500 en Pic (en cumul des sollicitation Florida + AS DSI)~~ 400 en nominal / 500 en Pic

En complément pour les parcours de consultation / résiliation d'offres :

Ressource/Service	Exposition	ST Source	Temps Interne (hors appel externe)	Temps global	Nombre d'appels/s supplémentaires
grappes-contrats grappes-formules-souscrites grappes-historiques-formules-souscrites	REFCLI	M3 Offre	20ms	30ms	~~100 en nominal / 250 en pic~~ 400 en nominal / 500 en Pic

Ce qui nous amène au global à 800 appels/s en nominal et 1000 en pics pour les grappes* et 400 en nominal / 500 en Pic pour le rechercherContexteClientIdentite

- [ ] why "offre incluse" info is not returned in the new API offres-additionnelles ?

Only "offre incluse" included in principal offer is not returned by the new API. According to DSI, there is no plan to retrieve this information without Change Request.

- [x] is it possible to have synchronous mode for subscribe and unsubscribe actions ?

No. There is a delay of max 1 second that shall be supported.

- [x] what is the relevency of "IncompatibleType" attribute returned by offres-additionnelles-souscriptibles while it is not documented in the related Swagger ?

This information shall not be used.

- [ ] error code expected for unsubscribing offer : an already unsubscribed offer or subscribable offer, an unsubscribing offer, a subscribing offer

- [ ] error code expected for subscribing offer : an already subscribed offer, an unsubscribing offer, a subscribing offer.

Error codes are decribed in DSI swagger (/contrats/{idContrat}/impact-souscription-offres-additionnelle). For example, when the user tries to subscribe an already subscribed offer, it will be returned as "incomatble offer".

- [ ] serviceTv attribute and unsubscribe action (maybe others) are not returned anymore by the API GET offres-additionnelles (no problem if the offerId is specified)

- [x] in offer unsubscribe API return, dateEffet et dateResiliationEcheance are returned ; in API GET offres-additionnelles return, there is only one attribute dateFin ⇒ what is mapped in dateFin ?

We shall use **dateResiliationEcheance** which is mandatory.

- [ ] During an offer subscription, what datas are updated by DSI for a related impacted offer with action "MODIFICATION" ?

- [x] shall we only use the HATEOS llinks returned by DSI or shall we build the query for subscribe/unsubscribe ?

We shall use DSI HATEOS links (validated by DSI with FG Filipe GOMES).

- [x] does a call on offres-additionnelles-souscriptibles with moderate ACR or basic ACR is expected to give the same result ?

Moderate ACR is planned to be used only for a transaction action (subscribe or unsubscribe) and shall not be used for other queries. However, after alignement with DSI teams (IL IMAD LAOUDINI), it has been agreed to use the same moderate token for other API in the context of a global subscribe/unsubscribe callflow.

- [x] Can the DSI model two offers with distinct commercialIds but the same servicesTv (one is an option and the other is a service offered) ?

That could happen.

DSI related issues

- [ ] During the tests, we noticed that one offer is both subscribed ("souscrite") and unsubscribed ("souscriptible").
  → to be fixed ?

- [ ] No subscription can be done : the command is acquitted but not applied (offer still unsubscribed)